BigDataAIHub Case Competition

Competition

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Problem & Constraints

Data Scale



5.9M transactions

High volume of data



≈61k customers

(53k individuals + 8k small businesses)

Challenges & Requirements



Labels are **scarce + noisy**



Evaluation is **ranking-focused** (Top-K)



Explainability required

Pipeline Overview



1. Data prep
customer-level master profile



2. Feature engineering
behavior signals



3. Embeddings
transaction sequence motifs



4. Supervised models
XGBoost rankers



5. Ensemble ranking
Combined risk scores



6. Unsupervised Isolation Forest
(optional blend)



7. Explanation + Rule mapping + Sources
Actionable insights

Data Prep: Master Customer Profile



merge customers ↔ transactions

handle missing occupation/industry/ /province

unify per-customer view

ensure consistent IDs between splits/features/ models

Feature Engineering (Tabular Behavior Signals)

Transaction & Activity Stats

- tx_count, active_days, amt_sum, amt_mean, amt_max, amt_std, amt_cv

Location & Channel Metrics

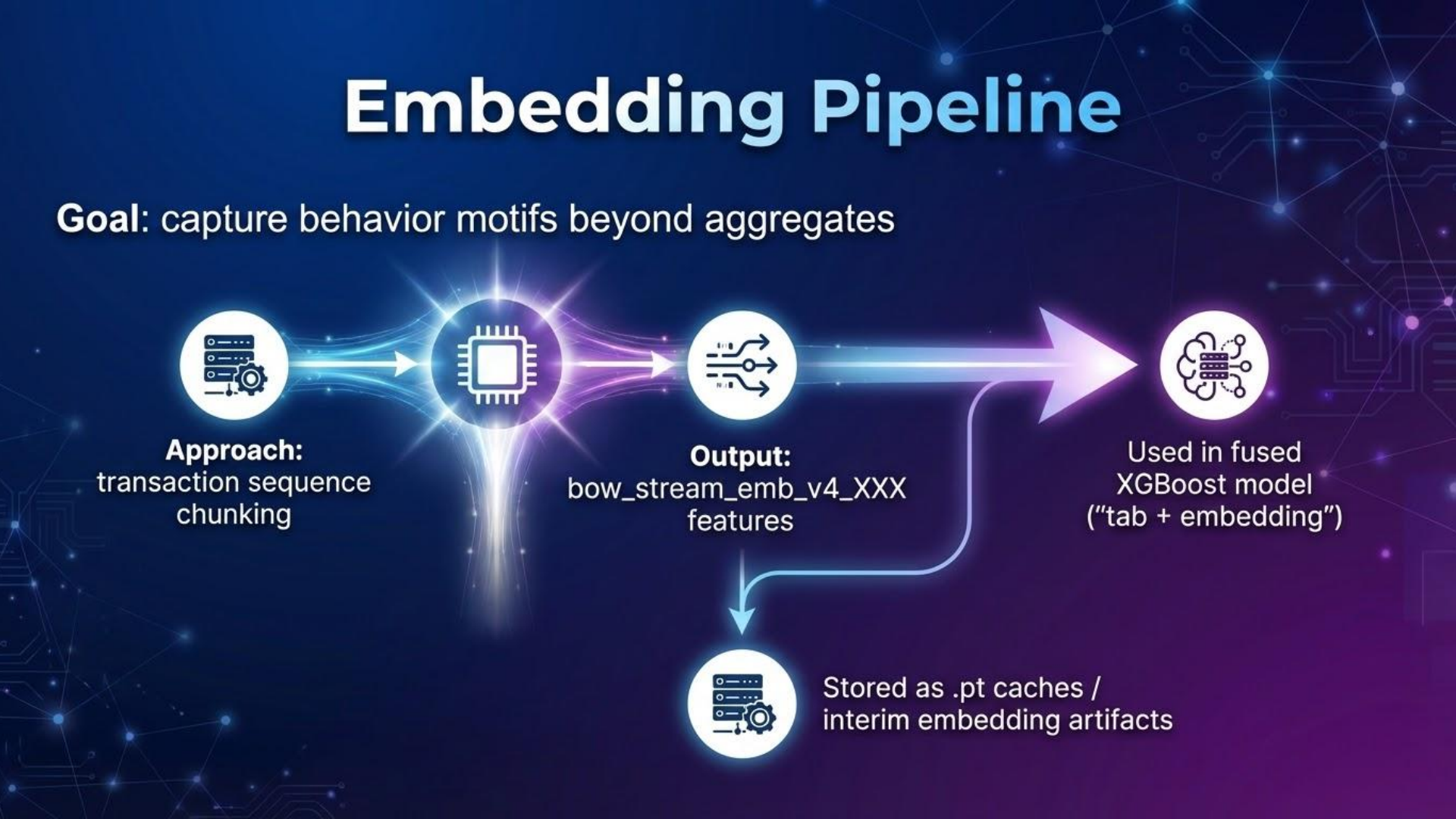
- night_ratio_all, intl_ratio_all, uniq_countries, uniq_cities
- uniq_channels, channel_entropy, uniq_mcc

Ratios, Deltas, & Flags

- debit_ratio, credit_ratio, cash_ratio
- deltas: *_first, *_last, *_delta
- smallbiz flags

Embedding Pipeline

Goal: capture behavior motifs beyond aggregates



Why ‘Tab + Embeddings’ Fusion Matters



Supervised Ranker: XGBoost (Fused Feature Model)

Model Overview & Process



Model Rationale & Details

Why XGBoost: top-tier tabular performance + native TreeSHAP for explainability

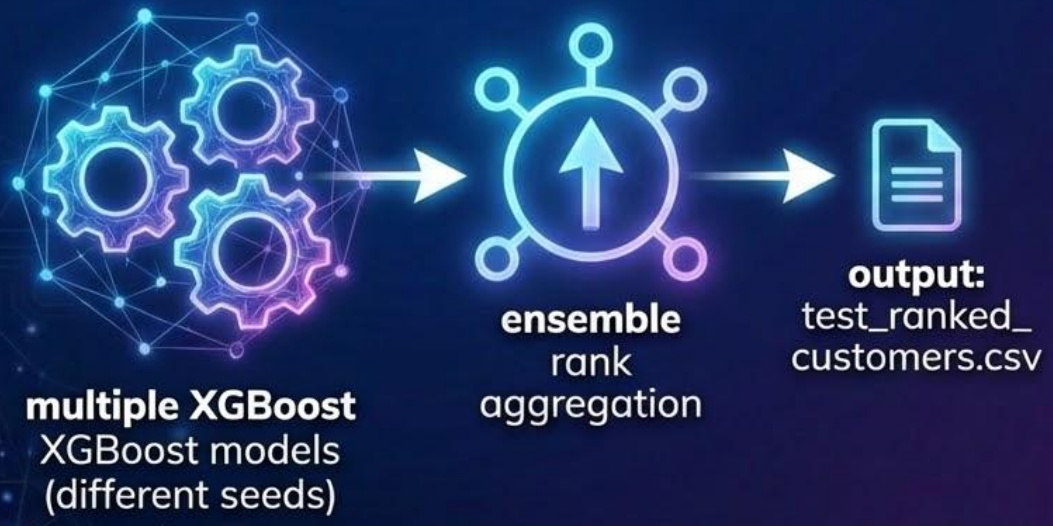
Imbalance handling: scale_pos_weight tuned to penalize missing positives

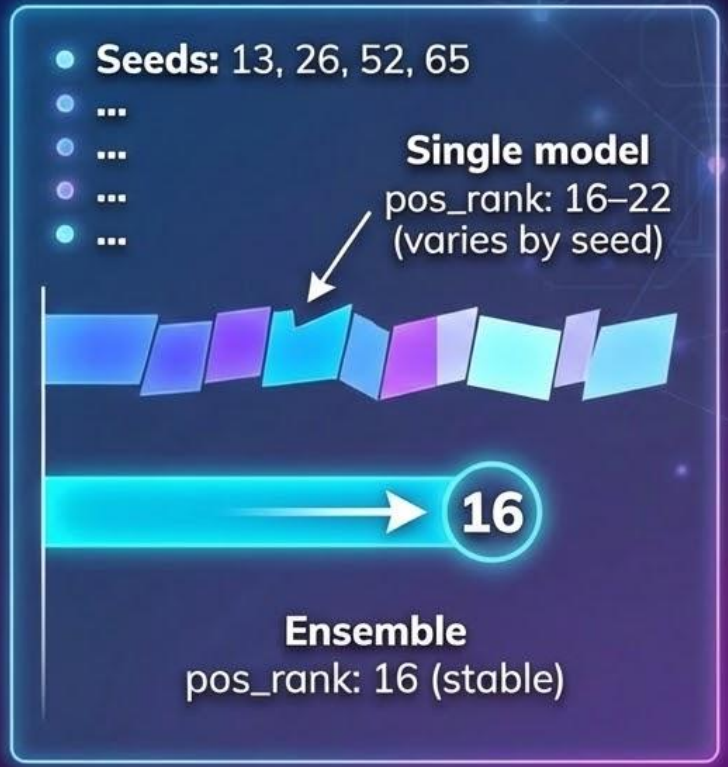
1
2
3
4
16

Evaluation metric: positive rank — rank 16 out of 61,000 customers

Labels: unlabeled customers treated as soft negatives (PU learning), not confirmed clean

Ensemble Ranking (Stability & Variance Reduction)





Unsupervised Learning (Isolation Forest)



Blending Strategy (Machine Learning Engineer View)

Final blending formula:

$$score_{blended} = (1 - \alpha) * score_{sup} + \alpha * score_{unsup}$$



Recommended α:

✓ **0.01 to 0.05**

✗ **not 0.85 / 0.95**

Explainability: What Investigators Actually Need



Score alone is useless



Need: reason codes tied to AML typologies

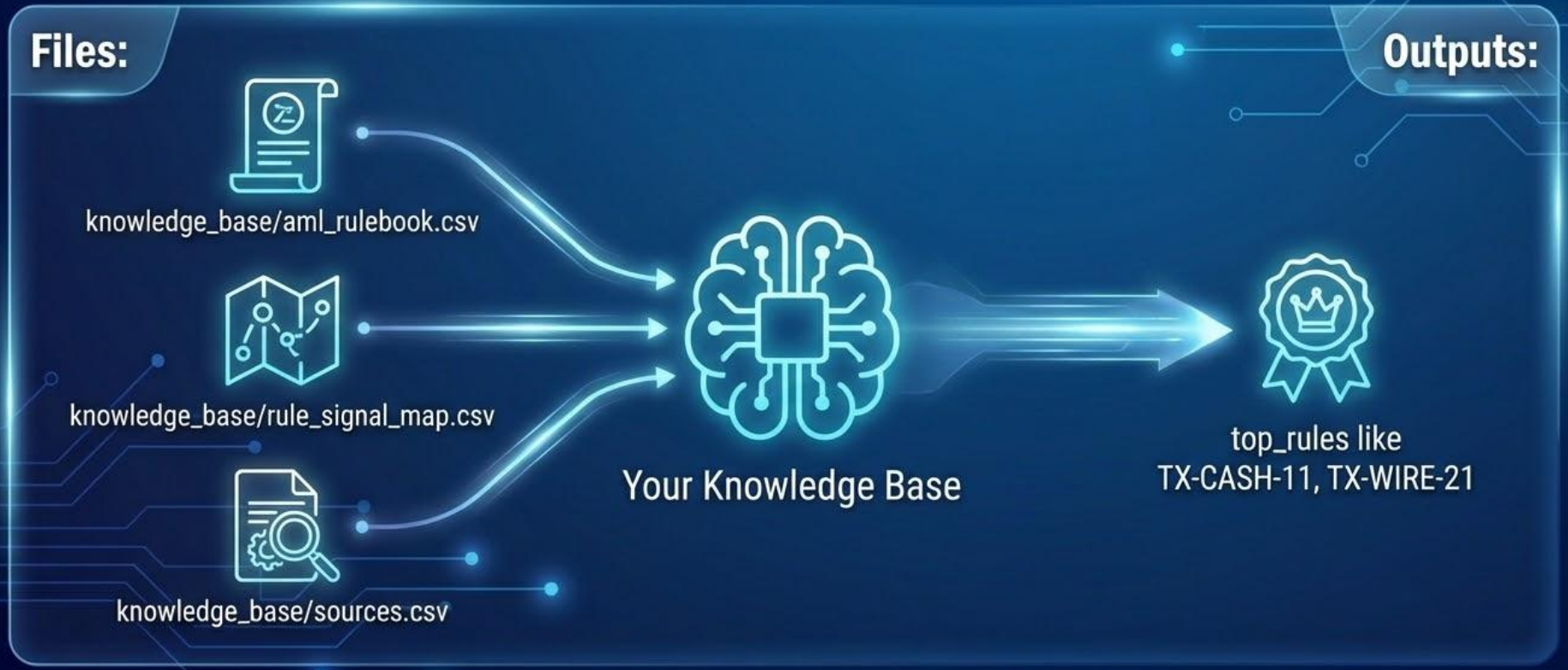


Need: evidence features and interpretable signals



Need: source links for credibility

Rulebook + Signal Map + Sources (Your Knowledge Base)



knowledge_base/aml_rulebook.csv

rule_id	category	red_flag	typology	source	feasibility
KYC-ID-01	KYC & Identity	Client provides information that is misleading	vague	hard to verify	low
KYC-ID-02	KYC & Identity	Common identifiers (addresses, phone numbers, IP) used by multiple apparently unrelated clients.	Network/CMLN	[1]	Medium
KYC-ID-03	KYC & Identity	Client occupation (e.g., student, unemployed, homemaker) is inconsistent with transaction volume or wealth.	CMLN / Opioids / General	[2], [3], [4]	Medium
KYC-ID-04	KYC & Identity	Client presents Chinese passport/visa with identical photos allegedly issued years apart.	CMLN	[5]	Low
KYC-ID-05	KYC & Identity	Client displays nervous behavior	avoids contact	or is defensive when	Low
KYC-ID-06	KYC & Identity	Sudden change in client's financial profile or pattern of activity.	General ML	[2], [7]	High
KYC-ID-07	KYC & Identity	Client opens accounts for several businesses for no apparent economic reason.	Opioids / General	[8]	Medium
KYC-ID-08	KYC & Identity	Accounts opened in the name of a foundation/association linked to suspected terrorist organizations.	Terrorist Financing	[9]	Low
TX-CASH-11	Transactions: Cash & ABM	High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts (Pass-through activity).	General / Opioids	[10], [8]	High
TX-CASH-12	Transactions: Cash & ABM	Structuring cash deposits at multiple branches/ABMs to avoid reporting thresholds (Smurfing).	General / CMLN	[11], [12]	High
TX-CASH-13	Transactions: Cash & ABM	Cash activity occurs between 10 p.m. and 6 a.m.	Human Trafficking	[13]	High
TX-CASH-14	Transactions: Cash & ABM	Frequent use of white-label ATMs (e.g., in hotels, convenience stores, casinos).	Human Trafficking	[14]	Medium
TX-CASH-15	Transactions: Cash & ABM	Trucking/Logistics employees making excessive cash deposits, particularly along known drug corridors.	Opioids	[15]	Medium
TX-CASH-16	Transactions: Cash & ABM	Large volumes of cash delivered via Armored Car Services (ACS) for a business with no apparent lawful purpose.	Bulk Cash Smuggling	[8], [16]	Medium
TX-CASH-17	Transactions: Cash & ABM	Large volume of cash delivered to US financial institution via ACS on behalf of a Mexico-based business.	Bulk Cash Smuggling	[17]	Low
TX-CASH-18	Transactions: Cash & ABM	Shipment of bulk cash delivered via ACS without accompanying instructions.	Bulk Cash Smuggling	[18]	Low
TX-WIRE-21	Transactions: Wires/EFT	Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without apparent business purpose.	Opioids / CMLN	[19], [20]	Medium
TX-WIRE-22	Transactions: Wires/EFT	Wires sent to chemical companies or brokers in China (Hebei/Hubei provinces), Eastern Europe, or India.	Opioids (Production)	[21], [20]	Low
TX-WIRE-23	Transactions: Wires/EFT	Large transfers between senders and receivers with no apparent relationship.	General / Human Trafficking	[22], [23]	Medium
TX-WIRE-24	Transactions: Wires/EFT	Wires routed through intermediary jurisdictions (Hong Kong, South Korea, Singapore) to conceal final destination.	Opioids / CMLN	[21], [24]	Low
TX-WIRE-25	Transactions: Wires/EFT	Client receives wires marked 'tuition' or 'living expenses' but funds are used for unrelated purposes (e.g., real estate, luxury goo	CMLN	[25], [5]	Low
TX-WIRE-26	Transactions: Wires/EFT	Mexico-based business transporting cash into US provides inconsistent info on currency originator in CMIR.	Bulk Cash Smuggling	[18]	Low
TX-EMT-31	Transactions: EMT (Email Transfer)	High volume of incoming EMTs rapidly depleted by outgoing EMTs to unrelated third parties.	Opioids / General	[26], [8]	High
TX-EMT-32	Transactions: EMT (Email Transfer)	EMTs containing references to drug slang (e.g., 'fent', 'china girl', 'dragon?s breath').	Opioids	[15]	Low
TX-EMT-33	Transactions: EMT (Email Transfer)	EMTs referencing 'escort', 'massage', or names of escort agencies.	Human Trafficking	[27]	Low
TX-EMT-34	Transactions: EMT (Email Transfer)	Multiple EMTs from unrelated third parties sent to the same beneficiary.	Human Trafficking	[28], [29]	High
TX-CARD-41	Transactions: Cards/Cheques	Excessive credit card payments or overpayments followed by cash advances.	General / Human Trafficking	[30], [31]	Medium
TX-CARD-42	Transactions: Cards/Cheques	Use of prepaid cards to do card-to-card transfers (e.g., male funding a female's card).	Human Trafficking	[32]	Low
TX-CARD-43	Transactions: Cards/Cheques	Use of bank drafts/cheques to purchase real estate or vehicles inconsistent with client profile.	Underground Banking	[33], [34]	Low
TX-CARD-44	Transactions: Cards/Cheques	Purchases at merchants selling gift cards/prepaid cards in rounded sums.	Human Trafficking	[35]	Medium
GEO-COR-51	Geographic/Corridors	Transactions along known drug corridors (e.g., Vancouver -> Calgary -> Saskatchewan; Toronto -> London/Hamilton).	Opioids	[36]	High
GEO-COR-52	Geographic/Corridors	Funds transferred to/from jurisdictions with weak AML controls or known for drug production.	General / TF	[19], [9]	Low
GEO-COR-53	Geographic/Corridors	Hotel/accommodation bookings in multiple cities over a short period.	Human Trafficking	[37]	Medium
IND-RE-61	Industry Specific: Real Estate	Incoming funds from China/HK used for structured draft purchases to developers referencing specific units/addresses.	Underground Banking	[38]	Low

knowledge_base/rule_signal_map.csv

rule_id	signal_id	entity	window	aggregation	columns_used	feasibility	description
KYC-ID-07	multiple_business_accounts	customer	all	count	customer_id; business_id	Medium	Customer linked to multiple businesses without clear economic rationale
TX-CASH-11	cash_to_wire_pass_through	transaction	3d	ratio	cash_indicator; channel; amount_cad; transaction_datetime	High	Cash deposits rapidly exiting via wire transfers
TX-CASH-12	cash_structuring_pattern	transaction	30d	count	amount_cad; transaction_datetime; branch_id	High	Multiple sub-threshold cash deposits across locations (structuring)
TX-CASH-13	night_cash_activity	transaction	30d	ratio	cash_indicator; transaction_datetime; amount_cad	High	High proportion of cash activity occurring between 10 p.m. and 6 a.m.
TX-CASH-14	white_label_atm_usage	transaction	30d	ratio	merchant_category; channel; amount_cad	Medium	Frequent use of non-bank or white-label ATMs
TX-CASH-15	logistics_worker_cash_volume	customer	30d	relative	occupation_code; amount_cad	Medium	Cash activity inconsistent with logistics or trucking occupation norms
TX-CASH-16	armored_car_cash_volume	transaction	30d	sum	channel; amount_cad	Medium	Large volumes of cash delivered via armored car services
TX-WIRE-21	high_risk_jurisdiction_wires	transaction	30d	ratio	country; amount_cad; channel	Medium	Wires involving high-risk jurisdictions without clear business purpose
TX-WIRE-23	unrelated_large_wire_transfers	transaction	30d	sum	amount_cad; counterparty_id	Medium	Large wire transfers between unrelated senders and receivers
TX-EMT-31	emt_pass_through	transaction	3d	ratio	channel; amount_cad; transaction_datetime	High	Incoming EMT funds rapidly depleted by outgoing EMTs
TX-EMT-34	multiple_emt_same_beneficiary	transaction	30d	count	channel; beneficiary_id	High	Multiple EMTs from unrelated third parties sent to the same beneficiary
TX-CARD-41	card_overpayment_cash_out	transaction	30d	ratio	channel; amount_cad; transaction_type	Medium	Credit card overpayments followed by cash advances
TX-CARD-44	rounded_gift_card_purchases	transaction	30d	count	merchant_category; amount_cad	Medium	Repeated purchases of gift or prepaid cards in rounded amounts
GEO-COR-51	drug_corridor_activity	transaction	30d	count	city; province; transaction_datetime	High	Transactions occurring along known drug trafficking corridors
GEO-COR-53	short_term_multi_city_hotels	transaction	14d	count	merchant_category; city	Medium	Hotel or accommodation payments across multiple cities in a short period
IND-AUTO-72	auto_trade_circular_flows	network	90d	cycle	industry_code; counterparty_id; amount_cad	Medium	Circular fund flows between auto dealerships and trade entities
IND-TRADE-81	trade_no_inventory_activity	customer	90d	ratio	industry_code; amount_cad; inventory_purchases	Medium	Trade businesses receiving wires without corresponding inventory purchases
IND-TRADE-82	trade_unrelated_jurisdictions	transaction	30d	ratio	country; industry_code; amount_cad	Medium	Trade businesses receiving funds from unrelated jurisdictions
IND-LEGAL-91	legal_trust_account_flows	transaction	30d	sum	merchant_category; amount_cad	Medium	Funds flowing to legal trust accounts without clear relationship
BEH-LIFE-111	escort_advertising_payments	transaction	30d	count	merchant_category; amount_cad	Medium	Payments linked to escort advertising or hosting services
BEH-LIFE-112	gambling_with_third_party_funds	transaction	30d	ratio	merchant_category; channel; amount_cad	High	Gambling activity funded by multiple third-party EMTs
KYC-ID-02	shared_identifier_network	network	all	count	address; phone; ip	Medium	Counts distinct customers sharing identifiers suggesting coordination
KYC-ID-03	txn_vs_occupation_peer	customer	30d	relative	occupation_code; amount_cad; income	Medium	Transaction volume anomalous relative to peers with same occupation
KYC-ID-06	financial_profile_change	customer	30d	delta	amount_cad; income; sales	High	Sudden change in transaction volume or financial profile versus historical baselinehases inconsistent with dec
BEH-LIFE-115	shipping_material_purchases	transaction	30d	count	merchant_category; amount_cad	Medium	Excessive purchase of shipping or mailing materials
TP-USE-121	nominee_behavior_pattern	customer	all	proxy	shared_identifier_network; txn_flow	Medium	Behavior indicative of acting on behalf of another party
TP-USE-122	daigou_fund_movement	transaction	30d	ratio	merchant_category; amount_cad; country	Medium	Daigou-style luxury purchases used to move funds

knowledge_base/sources.csv

Number	Red Flag / Indicator Rule	Corresponding Link(s)
1	Client provides information that is misleading, vague, hard to verify, or refuses to identify beneficial owners.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng
2	Common identifiers (addresses, phone numbers, IP) used by multiple apparently unrelated clients.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://www.fincen.gov/system/fil
3	Client occupation (e.g., student, unemployed, homemaker) is inconsistent with transaction volume or wealth.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://fintrac-canafe.canada.ca/i
4	Client presents Chinese passport/visa with identical photos allegedly issued years apart.	https://www.fincen.gov/system/files/2025-08/FinCEN-Advisory-CMLN-508.pdf
5	Client displays nervous behavior, avoids contact, or is defensive when questioned.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng
6	Sudden change in client's financial profile or pattern of activity.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng
7	Client opens accounts for several businesses for no apparent economic reason.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://fintrac-canafe.canada.ca/i
8	Accounts opened in the name of a foundation/association linked to suspected terrorist organizations.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng
9	High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts (Pass-through activity).	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://fintrac-canafe.canada.ca/i
10	Structuring cash deposits at multiple branches/ABMs to avoid reporting thresholds (Smurfing).	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://www.fincen.gov/system/fil
11	Cash activity occurs between 10 p.m. and 6 a.m.	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
12	Frequent use of white-label ATMs (e.g., in hotels, convenience stores, casinos).	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
13	Trucking/Logistics employees making excessive cash deposits, particularly along known drug corridors.	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
14	Large volumes of cash delivered via Armored Car Services (ACS) for a business with no apparent lawful purpose.	https://www.fincen.gov/system/files/shared/BCS-Alert-FINAL-508C.pdf
15	Large volume of cash delivered to US financial institution via ACS on behalf of a Mexico-based business.	https://www.fincen.gov/system/files/shared/BCS-Alert-FINAL-508C.pdf
16	Shipment of bulk cash delivered via ACS without accompanying instructions.	https://www.fincen.gov/system/files/shared/BCS-Alert-FINAL-508C.pdf
17	Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without apparent business purpose.	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng | https://www.fincen.gov/system/files/2025-08/FinCEN-Advisory-CMLN-508.pdf
18	Wires sent to chemical companies or brokers in China (Hebei/Hubei provinces), Eastern Europe, or India.	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
19	Large transfers between senders and receivers with no apparent relationship.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://fintrac-canafe.canada.ca/i
20	Wires routed through intermediary jurisdictions (Hong Kong, South Korea, Singapore) to conceal final destination.	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
21	Client receives wires marked 'tuition' or 'living expenses' but funds are used for unrelated purposes (e.g., real estate, luxury goods).	https://www.fincen.gov/system/files/2025-08/FinCEN-Advisory-CMLN-508.pdf
22	Mexico-based business transporting cash into US provides inconsistent info on currency originator in CMIR.	https://www.fincen.gov/system/files/shared/BCS-Alert-FINAL-508C.pdf
23	High volume of incoming EMTs rapidly depleted by outgoing EMTs to unrelated third parties.	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
24	EMTs containing references to drug slang (e.g., 'fent', 'china girl', 'dragon's breath').	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
25	EMTs referencing 'escort', 'massage', or names of escort agencies.	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
26	Multiple EMTs from unrelated third parties sent to the same beneficiary.	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
27	Excessive credit card payments or overpayments followed by cash advances.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng | https://fintrac-canafe.canada.ca/i
28	Use of prepaid cards to do card-to-card transfers (e.g., male funding a female's card).	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
29	Use of bank drafts/cheques to purchase real estate or vehicles inconsistent with client profile.	https://fintrac-canafe.canada.ca/intel/operation/ml-rec-eng.pdf
30	Purchases at merchants selling gift cards/prepaid cards in rounded sums.	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
31	Transactions along known drug corridors (e.g., Vancouver -> Calgary -> Saskatchewan; Toronto -> London/Hamilton).	https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
32	Funds transferred to/from jurisdictions with weak AML controls or known for drug production.	https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng
33	Hotel/accommodation bookings in multiple cities over a short period.	https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng.pdf
34	Incoming funds from China/HK used for structured draft purchases to developers referencing specific units/addresses	https://fintrac-canafe.canada.ca/intel/operation/ml-rec-eng.pdf

SHAP: What We Used



We did TreeSHAP contributions via XGBoost built-in: pred_contribs=True

Equivalent to SHAP-style additive attributions

Faster, no external SHAP dependency issues

Show Explainability Output Example

Customer ID: [SYNID_XXXX]

Blended Score: 0.9412

XGBoost Score: 0.9389 | **IsoForest Anomaly:** 94th percentile

Top Rules Triggered:

[R-ML-001] Large cash deposits inconsistent with profile

→ Cash Deposit Ratio: +0.18 | Source: FINTRAC ML Indicators

[R-ML-007] International wires to high-risk jurisdictions

→ Cross-Border Wire Volume: +0.14 | Source: FinCEN SAR Guidance

[R-TF-012] Transactions during unusual hours (2–4 AM)

→ Late-Night Ratio: +0.09 | Source: FINTRAC TF Indicators

Narrative:

"This customer exhibits concentrated late-night cash deposit activity across multiple ATMs, combined with significant cross-border wire transfers. This pattern is consistent with bulk cash smuggling or layering typologies as documented in FINTRAC and FinCEN guidance."

Explainability Actual Output (Top 3)

SYNID010543024 2	0.99107	1	0.1067	-0.16278	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without apparent business purpose. (driven by: Amt Sum; source: [19], [20])	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts (Pass-through activity). (driven by: Amt Max; source: [10], [8])	[TX-CASH-14] Frequent use of white-label ATMs (e.g., in hotels, convenience stores, casinos). (driven by: Intl Ratio All; source: [14])
SYNID020049249 7	0.79756	0.7959	0.96198	-0.11696	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts (Pass-through activity). (driven by: Channel Entropy; source: [10], [8])	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without apparent business purpose. (driven by: Uniq Cities; source: [19], [20])	[TX-EMT-31] High volume of incoming EMTs rapidly depleted by outgoing EMTs to unrelated third parties. (driven by: Marital Status Married; source: [26], [8])
SYNID010916459 4	0.64322	0.6432	0.64534	-0.1466	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts (Pass-through activity). (driven by: Channel Entropy; source: [10], [8])	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without apparent business purpose. (driven by: Intl Ratio All; source: [19], [20])	[TX-EMT-31] High volume of incoming EMTs rapidly depleted by outgoing EMTs to unrelated third parties. (driven by: Uniq Cities; source: [26], [8])

Implementation Choices & Engineering Tradeoffs



Results & What We Delivered

Deliverables:



Explanations CSV
(rules + sources)



Top Contribution CSV
(supervised + blended)



Scripts Reproducible
End-to-End

Top Contribution CSV

	customer_id	label	score	iforest_pct	score_sup	score_unsup	score_blended
2	SYNID0105430242	0	0.263529	0.1075232046897899	1.0	0.1066967062898263	0.9910669670628982
3	SYNID0200492497	0	0.20975128	0.9358817782120176	0.7958974716093623	0.9619768986330852	0.7975582658795995
4	SYNID0109164594	0	0.16951668	0.6292134831460674	0.6431951333773595	0.6453418968676968	0.6432166010122629
5	SYNID0106129658	0	0.14150017	0.4761602344894968	0.5368641002342366	0.4873144240630832	0.536368603472525
6	SYNID0100906564	0	0.107555136	0.388536069044296	0.408032545722328	0.3968424769238529	0.4079206450343432
7	SYNID0200771719	0	0.075387076	0.881875915974597	0.2859451389036355	0.9062158481429798	0.2921478459960289
8	SYNID0200875060	0	0.06741276	0.9689627096564076	0.2556802251477918	0.9961329589589252	0.2630847524859031
9	SYNID0103190083	0	0.061373185	0.1064973131411822	0.2327582320267579	0.1056374733090102	0.2314870244395804
10	SYNID0107928724	0	0.0536282	0.5506757857026543	0.2033636979920766	0.5642517275585521	0.2069725782877413
11	SYNID0200607748	0	0.0488265	0.9727080280084676	0.1851398106540986	1.0	0.1932884125475577
12	SYNID0107888998	0	0.046214145	0.4248656570591109	0.1752251422644026	0.4343527750222776	0.1778164185919813
13	SYNID0200018603	0	0.029292054	0.8648428594691419	0.1110007467571841	0.8886292180170487	0.1187770314697828
14	SYNID0200673561	0	0.018410392	0.9648265754763068	0.0697015855563686	0.9918624005918256	0.0789231937067232
15	SYNID0200525992	0	0.017726418	0.9420778374857516	0.0671056996996707	0.9683743295727762	0.0761183859984018
16	SYNID0100464450	0	0.018112978	0.3545839439830646	0.0685728104915504	0.3617869092254149	0.071504951478891
17	SYNID0103256605	0	0.017099548	0.6951473701351571	0.0647265405986111	0.7134186324125293	0.0712134615167502
18	SYNID0107832828	1	0.017835878	0.1153883732291157	0.0675211331311669	0.1148174924760832	0.0679940967246161
19	SYNID0105783650	0	0.014892736	0.6180263800683927	0.0563510291622244	0.6337912134102259	0.0621254310047044
20	SYNID0100075150	0	0.0139604965	0.5895945285784074	0.0528129014949372	0.6044353279418937	0.0583291257594068
21	SYNID0108220654	0	0.013874849	0.3618628887803289	0.0524878436215217	0.369302419422634	0.0556559893795328
22	SYNID0103024953	0	0.012130075	0.0833903273082559	0.0458659045569751	0.0817795114077771	0.0462250406254831
23	SYNID0200027516	0	0.009549905	0.9587363621560008	0.0360733878664633	0.9855742555946	0.0455683965437446
24	SYNID0106001660	0	0.010172247	0.4640612278130597	0.0384353618535287	0.4748222001782201	0.0427992302367756
25	SYNID0103249136	0	0.009287524	0.5952939260706726	0.0350775735124582	0.6103199556130943	0.0408299973334646
26	SYNID0200372275	0	0.0074552894	0.8600716495684742	0.0281236953178303	0.8837029439951579	0.0366794878046036
27	SYNID0109214068	0	0.0088721365	0.2058622374206155	0.0335010537378775	0.2082317534509138	0.0352483607350079
28	SYNID0106635947	0	0.008211094	0.3774303859306301	0.0309921997808175	0.3853758595759705	0.034536036378769
29	SYNID0105976167	0	0.0068023014	0.8753460348477446	0.0256454105967832	0.8994737461539756	0.0343836939523552
30	SYNID0107509503	0	0.0075113834	0.4411985018726592	0.0283365888214512	0.4512164366057468	0.0325653872992941
31	SYNID0100417825	0	0.006932614	0.5982250447809803	0.026139985880077	0.6133463355582831	0.0320120493775451
32	SYNID0101922420	0	0.006054963	0.8002442598925257	0.0228090378944185	0.821931166669469	0.030800259182169
33	SYNID0108713326	0	0.007865534	0.0746295391630027	0.029680696249045	0.0727339980160398	0.030111229266715
34	SYNID0104750690	0	0.0066302554	0.4789773652499592	0.0249924445803828	0.4902231114548481	0.0296447512491275
35	SYNID0101150580	0	0.0072733993	0.1081419964175215	0.0274333679982781	0.1073356087226995	0.0282323904055223
36	SYNID0108970515	0	0.006068066	0.3792216251424849	0.0228587676975694	0.3872253139869193	0.0265024331604629
37	SYNID0101861437	0	0.006567039	0.1415893177007002	0.024752519438972	0.1418699665416884	0.0259236939099991
38	SYNID0107403616	0	0.00471498	0.760022797589969	0.017723401718313	0.7804025085327102	0.025350192786457
39	SYNID0107633404	0	0.0055073956	0.4687184497638821	0.0207308558739779	0.479630781646687	0.025319855131705
40	SYNID0100663787	0	0.005858698	0.3394886826249796	0.022064153530883	0.346201052507692	0.0253055225206511
41	SYNID0103055630	0	0.005254676	0.5259729685718938	0.0197717094219924	0.5387460699093768	0.0249614530268663
42	SYNID0104253719	0	0.006563637	0.0416056016935352	0.0247396078316821	0.0386367839669116	0.0248785795930344
43	SYNID0106724091	0	0.004276083	0.7021006350757205	0.0160576563612569	0.7205978781713941	0.0231030585793582
44	SYNID0105519864	0	0.005682805	0.1613418010096075	0.0213965869991258	0.162264404727878	0.0228052651764133
45	SYNID0101898987	0	0.005021688	0.3897899364924279	0.0188874502922882	0.398137095011517	0.0226799467394805
46	SYNID0200093716	0	0.00356588	0.8954567659990229	0.0133622236063797	0.9202380752223548	0.0224309821225394
47	SYNID0105517063	0	0.00459982	0.4402051783097215	0.0172863350823629	0.4501908300687661	0.021615380032227
48	SYNID0109319576	0	0.0038711764	0.6740107474352711	0.0145209147362483	0.6915950703633338	0.0212916562925191
49	SYNID0100403533	0	0.0038391484	0.6539977202410031	0.0143993588990113	0.6709316206264608	0.0209646815162858
50	SYNID0200865648	0	0.0031512307	0.887575313466862	0.0117885055215407	0.9212004758141804	0.0207916252244671
51	SYNID0105518985	0	0.004439783	0.3697443413124898	0.0166789468119779	0.3774400188308086	0.0202865575321662

Explanation CSV

customer_id	score_blended	score_supervised	core_unsupervise	iforest_raw	reason_code_1	reason_code_2	reason_code_3
SYNID0105430242	0.99107	1	0.1067	-0.16278	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0200492497	0.79756	0.7959	0.96198	-0.11696	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0109164594	0.64322	0.6432	0.64534	-0.146	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0106129658	0.53637	0.53686	0.48731	-0.15179	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0100906564	0.40792	0.40803	0.39684	-0.15447	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0200771719	0.29215	0.28595	0.90622	-0.1261	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0200875060	0.26308	0.25568	0.99613	-0.10723	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0103190083	0.23149	0.23276	0.10564	-0.16282	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0107928724	0.20697	0.20336	0.56425	-0.1492	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0200607748	0.19329	0.18514	1	-0.10527	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0107888998	0.17782	0.17523	0.43435	-0.15339	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0200018603	0.11878	0.111	0.88863	-0.12826	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0200673561	0.07892	0.0697	0.99186	-0.10887	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0200525992	0.07612	0.06711	0.96837	-0.11553	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0100464450	0.0715	0.06857	0.36179	-0.15545	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0103256605	0.07121	0.06473	0.71342	-0.14258	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0107832828	0.06799	0.06752	0.11482	-0.1625	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0105783650	0.06213	0.05635	0.63379	-0.1465	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0100075150	0.05833	0.05281	0.60444	-0.14769	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0108220654	0.05566	0.05249	0.3693	-0.15525	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0103024953	0.04623	0.04587	0.08178	-0.16366	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0200027516	0.04557	0.03607	0.98557	-0.11103	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0106001660	0.0428	0.03844	0.47482	-0.15218	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0103249136	0.04083	0.03508	0.61032	-0.14747	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0200372275	0.03668	0.02812	0.8837	-0.12883	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0109214068	0.03525	0.0335	0.20823	-0.15971	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0106635947	0.03454	0.03099	0.38538	-0.15478	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0105976167	0.03438	0.02565	0.89947	-0.12695	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-CASH-14] Frequent use of white-label ATMs (e.g., in hotels, convenience stores, casinos). (driven	[TX-CASH-15] T
SYNID0107509503	0.03257	0.02834	0.45122	-0.15288	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0100417825	0.03201	0.02614	0.61335	-0.14735	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0101922420	0.0308	0.02281	0.82193	-0.13507	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig
SYNID0108713326	0.03011	0.02968	0.07273	-0.16401	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., Ch	[TX-CASH-11] High-volume cash deposits/withdrawals followed immediately by wire transfers or drafts	[TX-CASH-14] F
SYNID0104750690	0.02964	0.02499	0.49022	-0.1517	[TX-CASH-11] High-volume cash deposits/withdrawals followed imı	[TX-WIRE-21] Wire transfers to/from high-risk jurisdictions (e.g., China, Mexico, Colombia) without app	[TX-EMT-31] Hig

Financial Realities & Perceptions



The Request



The Brand Promise



The Demand



The Debt



The Stash